

Gene Pulser® Electroprotocol

Cell Type Fungal / Yeast

Molecules Electroporated DNA: *Aspergillus* genomic DNA, 3 to 13 kB.

Species Used *Aspergillus* spp.

Before the Pulse

Cell Growth Medium Czapek + requirement (ATCC#312) medium, Polypeptone -Dextrin medium.

Growth Phase at Harvest Protoplast

Pre-pulse Incubation None

Wash Solution 0.8 M Sorbitol

The Pulse

Instruments Used Gene Pulser® apparatus & Pulse Controller

Electroporation Temperature Room temperature, 25 °C

Electroporation Medium 1.1 M sorbitol

Cuvette Gap 0.4 and 0.2 cm

Cell Density 2 x 10⁷ / ml

Voltage Not given

Volume of Cells 2 x 10⁷ / ml

Field Strength 4 kV/cm

DNA Concentration 5 µg

DNA Resuspension Buffer TE buffer (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 25 µF

Volume of DNA Not given

Resistor (Pulse Controller) Not given.

After the Pulse

Time Constant 3 to 7 msec

Outgrowth Medium Czapek + 0.8 M Sorbitol

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature 30 °C

Length of Incubation 7 days

Selection Method or Assay Used Can grow on minimal medium

Electroporation Efficiency 10 / µg DNA

Per Cent Survival 10%

Name of Submitter Shigeomi Ushijima, Ph.D.

Institution Address Kikkoman Corp.
Research and Development Division
399 Noda
Noda City, Chiba 278 JAPAN

Survey Number

169